Class 22: Fixed Income II

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Today's Question

Silicon Valley Bank failed because of interest rate risk on “safe” Treasury bonds. What did they get wrong?

Positioning

Where We've Been	Where We Are	Where We're Going
Yield curve and term structure	Duration, immunization, and swaps	International investing

Matt Levine Reading

“Startup Bank Had a Startup Bank Run”

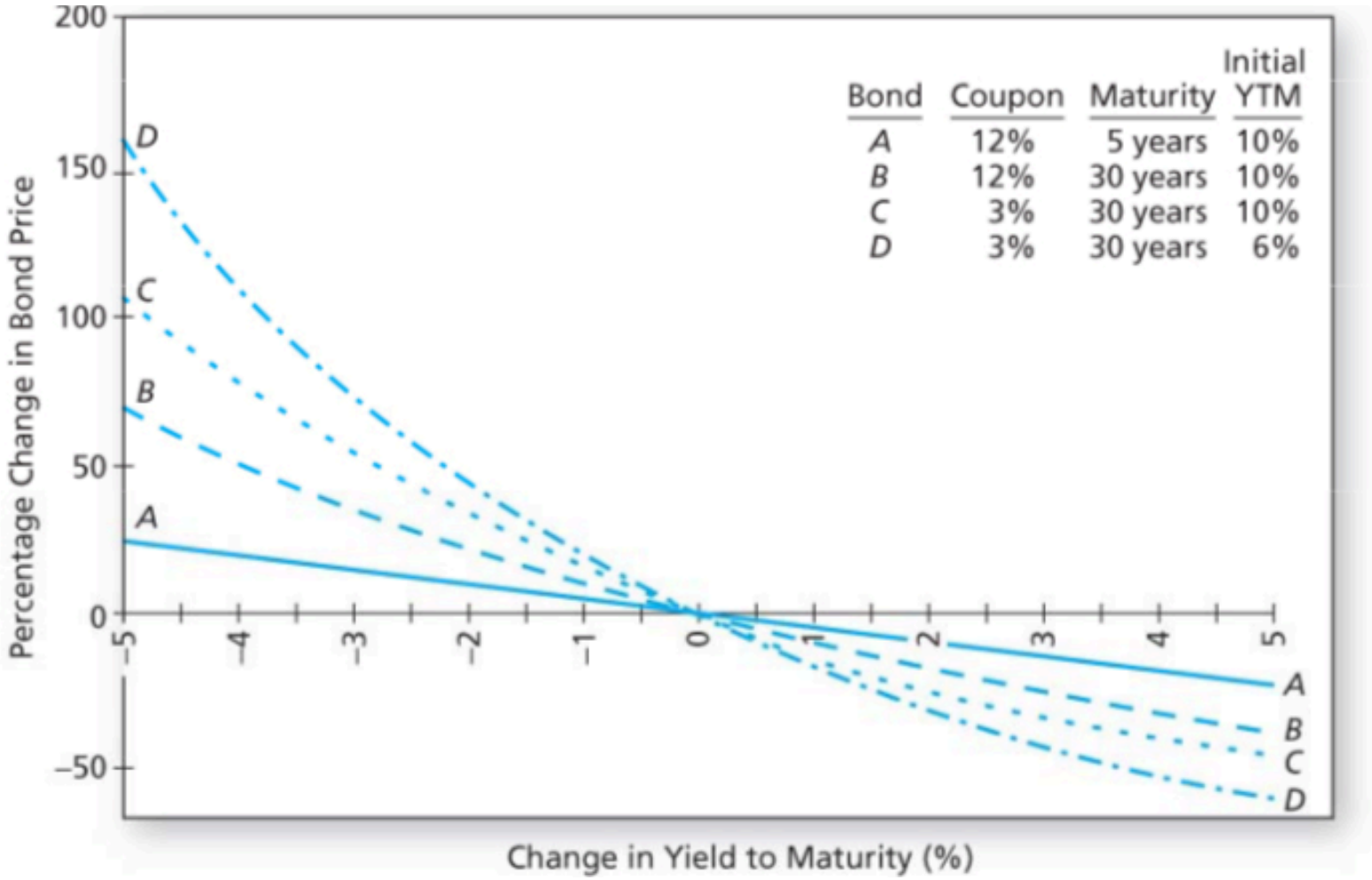
Key themes: held-to-maturity bonds, duration mismatch, and how a “safe” Treasury portfolio brought down a bank

Learning Objectives

By the end of today's class, you should be able to:

1. Describe the core bond-pricing relationships
2. Define and calculate duration and modified duration
3. Use duration to hedge a set of liabilities (immunization)
4. Explain how leveraged fixed-income positions can blow up (Orange County, SVB)
5. Describe the payoff structure of an interest-rate swap and why it provides "synthetic immunization"

Δ Bond Price and Δ Yield to Maturity



Bond Pricing Relationships

1. Bond prices and yields are inversely related.
2. Long-term bonds tend to be more price sensitive than short-term bonds.
3. Price sensitivity is inversely related to the bond's coupon rate.
4. Price sensitivity is inversely related to the yield to maturity at which the bond is selling.
5. An increase in a bond's yield to maturity results in a smaller price change than a decrease of equal magnitude.

Duration

- Need a single measure that tells us the interest rate sensitivity of a bond/portfolio
- **Duration** is a measure of the effective maturity
- The weighted average of the amount of time until each payment is received, with the weights proportional to the present value of the payment
- Duration is shorter than maturity for all bonds except zero coupon bonds
- Duration is equal to maturity for zero coupon bonds

Duration: Calculation

- Define duration as follows:

$$D = \sum_{t=1}^{T} t \times w_t$$

$$w_t = \frac{\left[\frac{\mathrm{CF}_t}{(1+\mathrm{YTM})^t}\right]}{P}$$

where YTM is the yield to maturity, P is the price, and CF_t is the cashflows at time t

- Can show that

$$\frac{\Delta P}{P} = -D\frac{\Delta(1+\mathrm{YTM})}{1+\mathrm{YTM}}$$

Modified Duration

- *Modified* duration (D^*) divides duration by $(1 + \mathrm{YTM})$:

$$D^* = \frac{D}{1 + \mathrm{YTM}}$$

- This gives a much cleaner sensitivity formula:

$$\frac{\Delta P}{P} = -D^* \Delta \ \mathrm{YTM}$$

- Why does this matter? $\Delta \ \mathrm{YTM}$ is what you see quoted on Bloomberg (e.g. "rates up 25 bps") — not $\Delta \ln(1 + \mathrm{YTM})$. So D^* is the number practitioners actually use.

Duration Matching (Immunization)

- Duration matters for investors who want to **immunize** themselves from interest rate risk
- Consider a pension fund with 30 years of obligations
  - Modified duration of liabilities ≈ 9.9 years (YTM = 7.75%)
  - Reflects the weighted-average maturity of the obligations
- A 1% **decrease** in interest rates raises the value of liabilities by $\approx 9.9\%$ — why?
- **Strategy:** hold an asset portfolio whose duration matches the liability duration

Duration Matching — Two Approaches

Possible immunizing portfolios:

1. A zero-coupon bond expiring in 9.9 years
2. A laddered portfolio of 25 zero-coupon bonds paying \$100K, \$103K, \$106K, ... in years 1, 2, 3, ...

Issues with these strategies?

- 9.9-year zero coupons may not exist at the right size
- Durations drift as time passes and as rates change — requires rebalancing
- Liability cash flows may themselves depend on rates (e.g. mortgage prepayments)

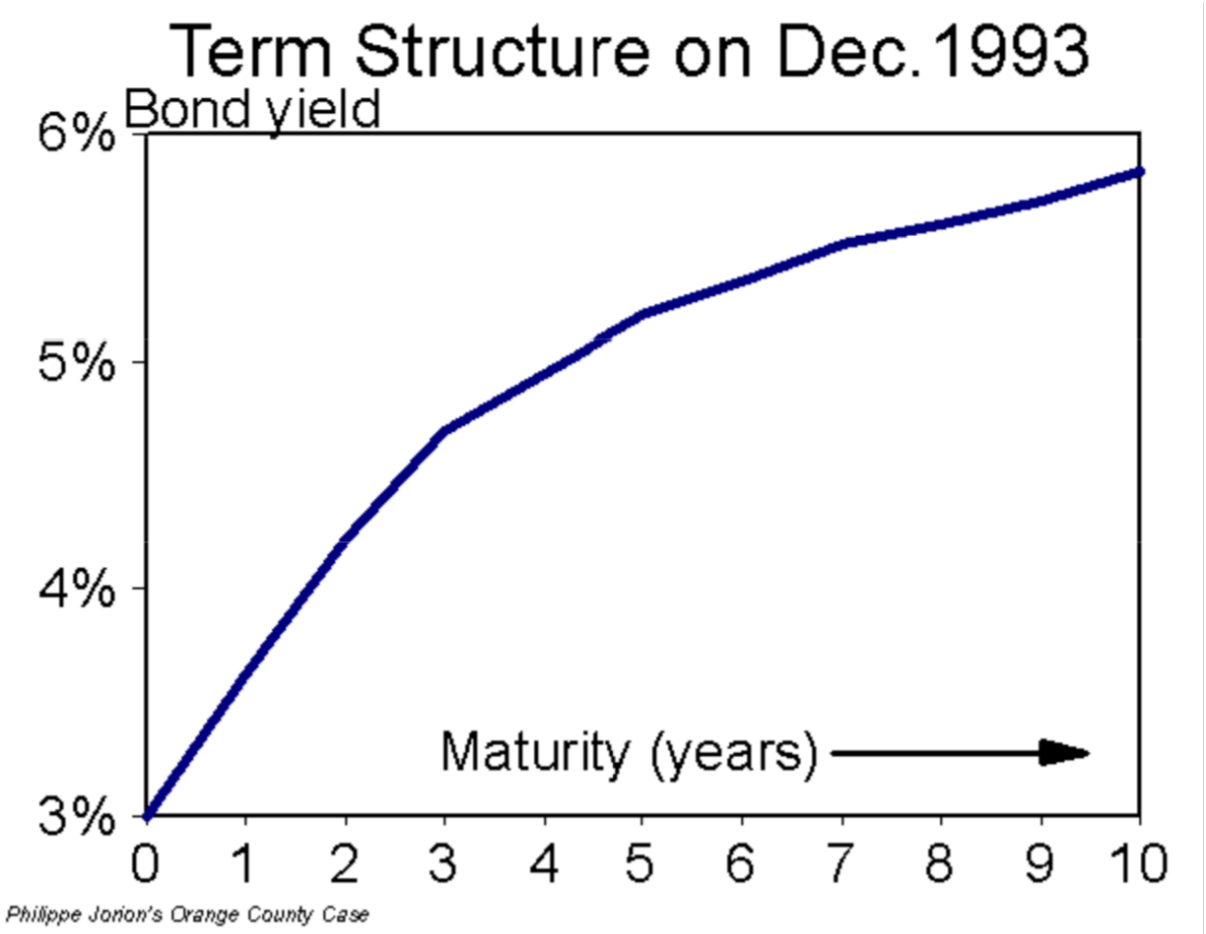
Duration: Important Caveats

- Duration is a **local approximation**
  - Not valid for large interest-rate changes
  - The true price–yield relationship is **convex** (Bond Pricing Relationship 4)

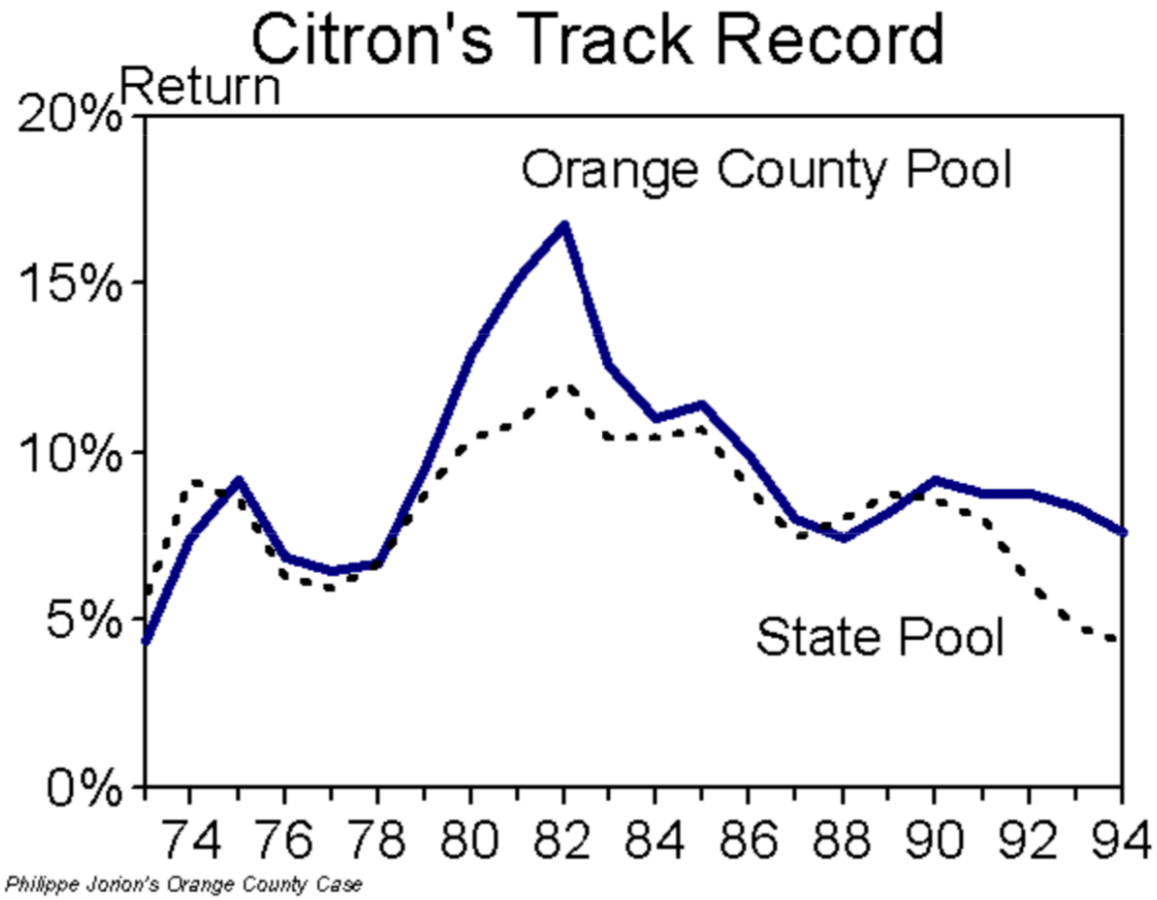
- Duration measures sensitivity to a change in **yield to maturity**
  - Implicitly assumes parallel shifts in a flat yield curve
  - Can be modified to accommodate non-parallel shifts (key-rate duration)

Application: Orange County, 1994

- Bob Citron managed Orange County funds as a large bet on interest rates
- **Strategy:** borrow heavily in short-term markets; reinvest in 5-year notes
- Via leverage, the portfolio's effective duration was 7.4 years
  - 2.74 (asset duration) $\times$ 2.7 (leverage)
- When the Fed hiked rates four consecutive times, the portfolio collapsed


Term Structure on Dec. 1993
Bond yield
6%
5%
4%
3%
Maturity (years)
0
1
2
3
4
5
6
7
8
9
10
Philippe Jorion's Orange County Case

Orange County



Orange County

Path of Interest Rates
5-Year Treasury yield
10%
9%
8%
7%
6%
5%
4%
90
91
92
93
94
Philippe Jorion's Orange County Case

Orange County: The Damage

- After four consecutive Fed rate hikes, Orange County posts a $1.6B loss on a $7.5B portfolio (pre-leverage)
- Declares bankruptcy — the largest municipal bankruptcy in US history at the time
- **Question:** how does a $1.6B loss compare to what an LP should have expected?
- Useful application of **duration** + **VaR** (value-at-risk)
- The same logic explains SVB in 2023: long-duration Treasuries + levered balance sheet + rate spike = equity wiped out

Interest Rate Swaps

- Fixed vs. floating interest rate swap
- One party pays a floating rate $\times$ notional value
- Other party pays a fixed rate (the **swap rate**) $\times$ notional value
- Exchanges are made periodically over a fixed term
- Fixed rate is set so that no cash changes hands at initiation
- Swaps provide **synthetic immunization**

Interest Rate Swaps: Example

		LIBOR	
	6.5%	7.0%	7.5%
7% X 100M	7M	7M	7M
LIBOR X 100M	6.5M	7M	7.5M
Net Swap Proceeds (fixed payer)	-.5M	0	.5M

- The counterparty that pays fixed and receives floating is effectively **short** (bonds)
- Combining a long bond portfolio with a short swap = synthetic floating-rate note

Pricing Swaps

- Consider a three-period fixed vs. floating rate swap
- Lending or borrowing at forward rates can be achieved synthetically, or in futures markets
- Consider cash flows for a hedged counterparty who receives fixed and pays floating:

t	Payoff on Forward Rate Agreement	Net Swap Payment	Net
1	-	$C-r_{0,1}$	$C-r_{0,1}$
2	$r_{1,2}-f_{1,2}$	$C-r_{1,2}$	$C-f_{1,2}$
3	$r_{2,3}-f_{2,3}$	$C-r_{2,3}$	$C-f_{2,3}$

Pricing Swaps

- C is set so that the net present value of these streams equals zero
- This is the **swap rate** — a weighted average of the spot and forward rates:

$$0 = \frac{C - r_{0,1}}{1 + r_1} + \frac{C - f_{1,2}}{(1 + r_2)^2} + \frac{C - f_{2,3}}{(1 + r_3)^3}$$

Next Class: April 28

- Topics: International Investing I
  - Benefits of international diversification
  - Home bias
  - Exchange-rate risk

- Matt Levine reading: TBD